

MBA Academic Advisor Bot

Hebrew University Business School

Client Documentation & Cost Analysis

Document Overview

- > System architecture & technology stack
- > Features & capabilities
- > Step-by-step usage guide
- > Full cost breakdown (current & scaled)
- > Future improvement roadmap

Prepared: May 2026

Technology: Python + Google Gemini AI

Current monthly cost: \$0

1. What Was Built

A web-based AI chatbot that acts as an academic advisor for students in the MBA program at the Hebrew University Business School. Students open a link in their browser on any device (phone, computer, tablet), type a question in Hebrew, and receive a precise answer based exclusively on official university sources.

The system was designed to reduce the volume of repetitive questions directed at the student secretariat, provide instant 24/7 responses, and ensure that all answers are grounded in verified, official academic information.

2. How It Works

Student types question >> Chat interface (website) >> Python server
>> Google Gemini AI (reads official HUJI content) >> Structured Hebrew answer

At every server startup, the system automatically fetches the latest content from all 6 official HUJI sources (web pages + PDF). This content is injected into the AI model's context. The AI is strictly instructed to answer only from this content - it cannot invent or guess information.

3. Official Sources

The bot reads and answers exclusively from the following 6 official HUJI sources:

Source	URL
Academic Regulations (PDF)	bschool.huji.ac.il - Academic Regulations PDF
Exemptions	bschool.huji.ac.il/ptorim
Admissions	bschool.huji.ac.il/mba/admittance
Specializations	bschool.huji.ac.il/mba/specializations
Accelerated Track	bschool.huji.ac.il/Accelerated-study-track
Accelerated Track - All Courses	bschool.huji.ac.il/Accelerated-study-track-all

4. Capabilities & Limitations

What the bot CAN answer:

- Admissions requirements and application process
- Available specializations and their structure
- Exemption requests - eligibility and process
- Accelerated study track - who qualifies, how to apply
- Academic regulations - credits, grades, course requirements
- General MBA program structure and rules

What the bot CANNOT answer:

- Questions not covered in the 6 official sources
- Personal student account issues (grades, registration systems)

- Real-time scheduling or current course availability
- Anything requiring a human administrative decision

In all unsupported cases, the bot directs the student to contact the student secretariat.

5. Features

Feature	Description
Hebrew UI	Full right-to-left interface, designed for Hebrew speakers
Source links	Sidebar with direct clickable links to all 6 official sources
Quick questions	One-click buttons for the 4 most common student topics
Chat history	Remembers context within a session for follow-up questions
Reset button	Start a fresh conversation at any time
Mobile friendly	Fully responsive - works on phones and tablets
Disclaimer	Every answer includes a reminder to verify with secretariat
Typing indicator	Shows animation while the AI is generating a response
Timestamps	Each message shows the time it was sent
Error handling	Clear messages if the server or API is unavailable

6. Technology Stack

Component	Technology	Reason
Backend	Python + FastAPI	Fast, reliable, easy to maintain
AI Model	Google Gemini 2.5 Flash	Best Hebrew support, fast responses
Content loader	BeautifulSoup + pdfplumber	Fetches real content from HUJI at startup
Frontend	HTML + CSS + JavaScript	No framework needed, loads instantly
Web server	Uvicorn	Industry-standard Python ASGI server
Sharing	ngrok	Instant public URL for local server

7. How to Use - Step by Step

Requirements

- A computer with Python 3.10 or higher installed
- Internet connection (to reach HUJI sources and Gemini API)
- Google Gemini API key (already configured in the project)

Starting the server

Step	Action
1	Open a terminal (Command Prompt or PowerShell)
2	Navigate to project: <code>cd "C:\...\final_project_big"</code>
3	Run: <code>uvicorn app:app --reload</code>

4	Wait for: [startup] Done. Total content: ~31000 chars
5	Open browser at: http://localhost:8000

Sharing with students (ngrok)

- Keep uvicorn running in the first terminal
- Open a second terminal and run: ngrok http 8000
- Copy the public link (e.g. https://abc123.ngrok-free.app)
- Send the link to students via WhatsApp, email, or the university portal
- Note: The link changes every time ngrok restarts. The computer must stay on.

Stopping the server

Press CTRL + C in the terminal window where uvicorn is running.

8. Cost Breakdown

Current Setup - Local + ngrok (running today)

Item	Plan	Monthly Cost
Google Gemini 2.5 Flash API	Free tier (1,500 req/day)	\$0
ngrok	Free tier (1 public URL)	\$0
Hosting	Local computer	\$0

Total current monthly cost: \$0

Limitation: computer must stay on and connected for the bot to work.

Cloud Deployment (recommended for full student access - 24/7)

If the university wants the bot always online for all students:

Item	Option	Monthly Cost
Cloud hosting	Railway.app (Starter)	~\$5
Cloud hosting	Render.com (free tier)	\$0 (limited)
Cloud hosting	Google Cloud Run	\$5-\$15
Domain name (optional)	Custom URL	~\$1-\$2
Gemini API (free tier)	Up to 1,500 req/day	\$0
Gemini API (paid)	Above free tier	~\$0.15/1M tokens

Estimated cost for ~50 students/day: \$5-\$10 / month

Includes hosting + Gemini API at moderate usage.

Gemini API Pricing Detail

Model	Input tokens	Output tokens
Gemini 2.5 Flash (free)	1,500 req/day free	1,500 req/day free
Gemini 2.5 Flash (paid)	\$0.15 per 1M tokens	\$0.60 per 1M tokens

Each student conversation uses approx. 5,000-8,000 tokens (full source content is included in every request). At paid rates: 100 conversations = approx. \$0.05-\$0.08 total.

9. Important Notes

Content refresh

Content is fetched at startup only. If HUJI updates their website, restart the server to reload.

Not a replacement

The bot supplements - not replaces - the student secretariat. Every answer includes a disclaimer.

Session memory

Conversation history is temporary and lost when the server restarts.

API key security

The .env file contains the API key. Never share it or upload it to GitHub.

PDF parsing

The academic regulations PDF is parsed automatically. A server restart reloads updated content.

10. Future Improvements & Roadmap

Feature	Effort	Est. Cost
Deploy to cloud (always online, no local computer needed)	Low	~\$5/month
WhatsApp / Telegram integration	Medium	Provider cost
Admin panel to update sources manually	Medium	Dev cost only
Student login and personalization	High	Dev cost only
Analytics dashboard (topics, usage, frequency)	Medium	Dev cost only
Persistent chat history per student	Medium	~\$0 DB cost
Multi-language support (English)	Low	Dev cost only

11. Contact & Support

For technical questions or to request upgrades, contact the development team.

For content accuracy questions, contact the MBA program secretariat at the Hebrew University Business School.

This document was prepared at the end of the initial development phase - May 2026.

All costs are estimates based on current provider pricing and may change.